

Assignment - 4

Section - A

1. What is Software Configuration Management?

Ans) Software Configuration Management (SCM) is a process of systematically controlling, organizing & managing changes in software products throughout software development life cycle. It ensures consistency, version control, & integrity of project artifacts like code, documents and designs.

2. What are project specification parameters?

Ans) It includes:-

- Scope - Defines Objectives & Deliverables.
- Time - Schedules & Deadlines
- Cost - Budget allocation
- Quality - Quality standards to be achieved.
- Resource - Team, tools, hardware.
- Risk - Identifying uncertainties & mitigation.

3. What do you mean by Project Management?

Ans) Project Management is the process of planning, organizing, and executing and controlling projects activities to achieve goals within time, cost, scope and quality constraints.

4. List of the phases of the project life cycle.

Ans) i) Initiation

ii) Planning

iii) Execution

iv) Monitoring & Controlling

(v) Closing

5. What is scope creep, and how does it lead to project delays?

Ans) Scope creep refers to uncontrolled changes or addition of new features to a project without adjusting time or resources. It causes delays by:

- Increasing workload.
- Requiring extra resources.
- Causing nervousness & confusion.

Section - B

1. State the purpose of SCM. Explain activities in SCM.

Ans) → Purpose of SCM:-

Purpose	Explanation
• Control Changes	Prevent uncontrolled modifications, manage what & how changes occur.
• Maintain Integrity	Ensures software consistency across versions.
• Version Management	Track and store revisions of code and documents.
• Improved team collaboration	Helps multiple developers work together without conflict.
• Audit & Compliance	Ensures standards & documented change control.
• Reduce errors	Minimize configuration conflicts & human errors.

→ Activities in SCM:-

SCM Activities	Description
• Configuration Identification	Identify items to be controlled (code, docs, test cases).
• Version Control	Maintain history of changes; track revisions.
• Change Control	Analyze, review & approve change requests.
• Configuration Status Accounting	Recording & reporting status of configuration items.
• Configuration Audits	Verify that standards and approved changes are followed.
• Build & Release Management	Manage build processes and release versions.

2. Explain Management spectrum 4 P's in software project planning.

Ans) i) People :-

People are the most valuable asset in software engineering.

→ Includes :

- Project Managers
- Developers
- Testing ^{exp}
- Stakeholders
- Users

→ Key Areas :

- Team Capability & training
- Communication
- Leadership style

ii) Product :-

Understanding product before development begins.

→ Activities:

- defines system objectives.
- Understand scope, features, constraints.
- Identify business needs & users.

→ Document Created

- SRS (Software Requirement Specification)
- Project Scope Statement.

iii) Process :-

Defines structured Methodology for developing software.

→ Includes :-

- SDLC model (waterfall, agile, spiral)
- Tools & technologies
- Quality assurance & testing strategies

iv) Project :-

Project includes planning, scheduling, monitoring, & controlling tasks.

→ Key Components:

- Work Breakdown structure (WBS)
- Gantt Chart / PERT Chart
- Budget Planning
- Risk Management
- Milestones & deliverables

3. Describe the process of creating a project management plan with example.

Ans) A project management plan (PMP) defines how a project will be executed, monitored & completed. It serves as a roadmap for entire project.

→ Steps for Creating PMP

Step	Description	Example
i) Project Objectives	Define clear goals & success criteria	Build online learning platform.
ii) Scope Definition	Identify deliverables, boundaries & exclusions.	Course videos, quizzes, user login.
iii) Work Breakdown Structure	Break work into manageable tasks	UI → login Page → Dashboard → Course module
iv) Schedule & Timeline	Use PERT/CPM or Gantt Chart	6 month project plan.
v) Resource Allocation	Assign team, tools & equipment.	1 PM, 2 developers, 1 Tester.
vi) Cost & Budget Estimation	Estimate expenses	Development cost ₹4,00,000.
vii) Risk Management	Identify risks & plan mitigation.	Server failure → Cloud backup.
viii) Communication Plan	Reporting mechanism	Weekly review meeting, slack updates
ix) Quality Management	Standards, testing, quality checks	ISO quality standards, automated testing.

Section - C

- Describe the COCOMO model, including Basics, Intermediate & detailed versions. Include a real-world example with effort, time & staffing calculations.

Ans)

COCOMO (Constructive Cost Model) is a software cost estimation model introduced by Barry Boehm.

It predicts:

- Software development effort (person-months).
- Time required to complete the project.
- Team size (staffing).

It is based on size in KLOC (Kilo lines of code).

→ Types of COCOMO Model

Model Type	Description	Usage
• Basic COCOMO	Provides rough cost & time estimation	Small software projects.
• Intermediate COCOMO	Considers cost drivers along with size.	Medium scale system.
• Detailed COCOMO	Phase-wise calculation + cost drivers	Large, complex systems.

→ COCOMO Formula:-

- $\text{Effort (PM)} = a \times (\text{KLOC})^b$
 - $\text{Development Time (T)} = c \times (\text{Effort})^d$
 - $\text{Team size} = \text{Effort} / \text{Development Time}$
- Values of a, b, c, d vary by project type.

Software Type	a	b	c
Organic	2.4	1.05	2.5
Semi-Detached	3.0	1.12	2.5
Embedded	3.6	1.20	2.5

2. Differentiate b/w PERT and CPM. Also describe the CPM technique with an example.

Ans)	Feature	PERT	CPM
•	Full Form	Program Evaluation Review Technique	Critical Path Method
•	Focus	Time Estimation	Cost & Time
•	Type	Probabilistic	Deterministic
•	Used for	R&D, Research Projects	Construction, software, industrial projects
•	Time Estimate	Optimistic, Pessimistic, most likely	Fixed duration
•	Nature	Event-oriented	Activity-oriented

→ Critical Path Method (CPM)

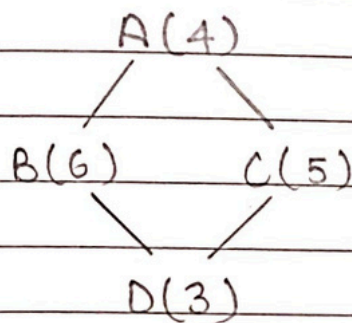
A project scheduling technique used to determine the longest path from start to finish, known as critical path.

→ Steps in CPM

Step	Description
1	List all activities with dependencies
2	Draw network diagram
3	Calculate ES, EF
4	Calculate LS, LF
5	Determine slack/float
6	Identify critical path

→ Numerical Example (Simple)

Activity	Duration	Predecessor
A	4	-
B	6	A
C	5	A
D	3	B, C



Paths

- $A \rightarrow B \rightarrow D = 4 + 6 + 3 = 13$
- $A \rightarrow C \rightarrow D = 4 + 5 + 3 = 12$

critical Path = $A \rightarrow B \rightarrow D$

Project Duration = 13 units

3. Solve the following according to Critical Path Method (CPM).

Activity	Predecessor	Duration (weeks)
A	-	3
B	A	6
C	A	4
D	B	5
E	B, C	3
F	C	2
G	D, E, F	4

a) Draw the network diagram for project.

b) Find earliest start time (ES) & latest finish time (LF).

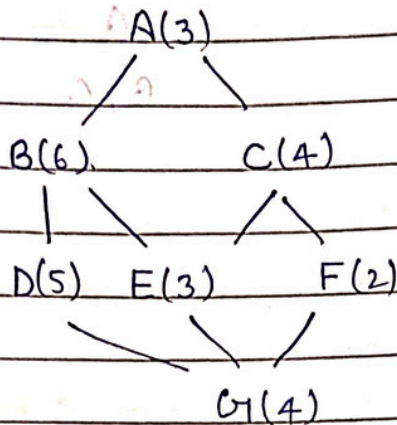
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- c) Identify critical path & total project duration.
 d) Find Total Float (Slack) & Free Float of project.

Ans) a)



Forward Pass

b)	Activity	Duration	ES	EF
	A	3	0	3
	B	6	3	9
	C	4	3	7
	D	5	9	14
	E	3	9	12
	F	2	7	9
	G	4	14	18

(Answer) Total Project Duration = 18 weeks.

c) Backward Pass

Activity	Duration	LS	LF
G	4	14	18
D	5	9	14
E	3	11	14
F	2	12	14
B	6	3	9
C	4	7	11
A	3	0	3

Slack Calculation

$$\text{Slack} = \text{LF} - \text{EF}$$

Activity	ES	EF	LS	LF	Slack
A	0	3	0	3	0
B	3	9	3	9	0
C	3	7	7	11	4
D	9	14	9	14	0
E	9	12	11	14	2
F	7	9	12	14	5
G	14	18	14	18	0

- Critical Path - Activities with 0 slack

A → B → D → G

(d) Total Float & Free Float

Activity	Total Float	Free Float
A	0	0
B	0	0
C	4	0
D	0	0
E	2	0
F	5	0
G	0	0

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17/11/25